

SYED MUHAMMAD HUZAIFA

AI Engineer | Full-Stack Developer | Automation Specialist

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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with hands-on experience delivering production AI systems, SaaS platforms, and automation pipelines. Currently building agentic AI, LLM-powered chatbots, full-stack web applications, and workflow automation tools at a technology company. Former AI Applied Research Engineer at QLU.ai. Published researcher in audio similarity for voice clone detection. 3rd place in two national AI competitions. FAST NUCES Computer Science graduate (2025), Dean's List.

TECHNICAL SKILLS

AI / ML / GenAI: LLMs · RAG · Agentic AI · QLoRA Fine-tuning · LangChain · LangGraph · OpenAI GPT-4o · Claude (Anthropic) · Gemini · Hugging Face · PyTorch · TensorFlow · pgvector · YOLO · Computer Vision · NLP · Transformers · Whisper · Prompt Engineering

Backend: Python · FastAPI · Node.js · Express · Fastify · PHP · Laravel · Cloudflare Workers · REST APIs · WebSockets · SSE · BullMQ · Redis Queues · APScheduler · Nginx

Frontend & Mobile: React 18 · Next.js 14 · Vite · TypeScript · JavaScript · React Native · Expo · Tailwind CSS · shadcn-ui · Radix UI · Vue.js 3 · HTML/CSS

Databases: PostgreSQL · pgvector · Redis 7 · MongoDB · MySQL · SQLite · Supabase · Firebase

DevOps & Infra: Docker · Docker Compose · AWS EC2 · Vercel · GitHub Actions · CI/CD · Bitbucket Pipelines · Playwright · Vitest · Prometheus

Integrations: HubSpot · Stripe · Booqable · EverWebinar · WhatsApp Web.js · Apollo.io · Apify · Slack · Webhooks

PROFESSIONAL EXPERIENCE

AI & Full-Stack Developer · *Technology Company* · Karachi, Pakistan

Mar 2026 – Present

Building production AI systems, full-stack web applications, and automation pipelines across diverse client projects.

- **AI Lead Automation Platform:** Built a fully autonomous lead outreach system (TypeScript monorepo) — Apollo.io scraping via Apify, GPT-4o email personalization, BullMQ Redis job queue, real-time Next.js 14 + Recharts dashboard, Docker Compose full stack, GitHub Actions CI/CD.
- **Multi-Tenant AI Chatbot SaaS (Chatty Bot):** Architected a full SaaS platform — React 18 + Vite frontend, Fastify backend, PostgreSQL + pgvector RAG knowledge base, Stripe billing, embeddable widget.js, Bull Board queue dashboard, Playwright E2E testing.
- **WordPress AI Chatbot (Cloudflare Workers):** Deployed Claude-powered chatbots on Cloudflare Workers embedded into WordPress sites — full HubSpot CRM lead capture, CORS/rate limiting, abuse protection, mobile-responsive UI.
- **Golf Club Rental Platform (FireStx):** Delivered full-stack booking platform — React + Vite on Vercel, Node.js/Express on AWS EC2, MongoDB, Booqable real-time inventory API, 8-step rental wizard, Bitbucket Pipelines CI/CD with test and production environments.
- **EverWebinar AI Bot:** Built Python + Playwright bot joining webinar sessions, classifying 50+ attendee intent categories via Claude (few-shot learning), responding with contextual cooldowns. AWS EC2 deployment automated via GitHub Actions.
- **WhatsApp Business Platform:** Developed full-stack platform combining real-time WhatsApp message management (QR-based), invoice creation/tracking, and Recharts analytics — React + Vite + Express + SQLite + whatsapp-web.js.
- **Email Automation Service:** Built multi-account Gmail IMAP watcher with domain allowlist filtering, GPT-powered AI replies per business profile, SMTP delivery, and SQLite deduplication across restarts.

AI Applied Research Engineer · *QLU.ai*

Aug 2025 – Feb 2026

- **Deep Search Optimization:** Improved profile ranking precision and recall using agent-based semantic matching techniques for the company's core AI product.
- **LLM Fine-tuning:** Trained and fine-tuned QLoRA-based document ranking models — delivering faster, more relevant retrieval compared to traditional search pipelines.
- **Agentic Search System:** Architected a Perplexity-style search experience using Gemini 2.5 Flash + Serper with agentic subquery decomposition, cached multi-session summaries, and real-time SSE answer streaming.
- **Voice Cloning Pipeline:** Designed real-time voice cloning system — Chatterbox + Silero VAD + Whisper transcription, active interruption handling, SSE streaming, ~5s average end-to-end latency.
- **Published Research:** Co-authored paper accepted at AIRC: 'Multi-stream Perceptual Audio Similarity for Voice Clone Detection.' Engineered Multiscale Temporal Fusion architecture — 23.5% improvement in Pearson correlation (0.5909) over baseline.

Junior Full-Stack Developer · [Alahdeen](#) · Karachi, Pakistan

Jun 2025 — Aug 2025

- Built core e-commerce features: advanced product filtering, company profile pages, and customer review & rating system — improving discovery and engagement.
- Developed administrative portal modules: user management and role-based access control (ACL).
- Full-stack delivery using React (frontend), PHP (backend), Git source control, and Hostinger deployment.

Software Engineer Fellow · [HeadStarter AI](#)

Jul 2024 — Sep 2024

- Built a production RAG-powered chatbot, Pantry Tracker web app, and AI-powered portfolio — shipped across the 7-week fellowship.

Machine Learning Intern · [InternCareer](#)

Aug 2024 — Sep 2024

- Developed and evaluated machine learning models as part of a 2-month remote internship program.

SELECTED PROJECTS (AI/ML RESEARCH)

AI-Based Diabetic Foot Ulcer Management — Final Year Project · [React Native](#) · [FastAPI](#) · [PyTorch](#) · [Supabase](#)

- React Native mobile app with deep learning DFU staging, heatmap visualization, and similar-case image retrieval. Reduced hospital dependency for DFU patients in Pakistan through early detection.

Video Anomaly Detection — CNN-LSTM on UCF-Crime · [PyTorch](#) · [Computer Vision](#) · [TPU](#)

- Hybrid CNN-LSTM with attention mechanism — 74.8% accuracy across 14 anomaly classes. 40% faster training via TPU acceleration. Triplet loss embeddings, t-SNE visualization.

Fingertip Detection & Fingerprint Extraction — PaysysLab 3rd Place · [YOLO v11](#) · [OpenCV](#) · [Biometrics](#)

- YOLO v11 pipeline for accurate fingertip detection and biometric fingerprint extraction from palm images.

Sonography Image Segmentation — Procom CV Competition 3rd Place · [PyTorch](#) · [Medical Imaging](#)

- Deep learning segmentation model for post-surgery pain management using sonography images.

PASCAL VOC 2007 Object Detection · [TensorFlow](#) · [Transfer Learning](#)

- Multi-task learning with MobileNetV2 backbone — 70% classification accuracy, 0.75 mIoU with focal loss for class imbalance.

EDUCATION

Bachelor of Science in Computer Science · [FAST NUCES](#) · Karachi, Pakistan

Aug 2021 — Jun 2025

Honors: Dean's List · Relevant: Deep Learning, Computer Vision, AI, Data Science, DevOps, Software Engineering, Information Retrieval, App Development

CERTIFICATIONS & ACHIEVEMENTS

- Published Research — 'Multi-stream Perceptual Audio Similarity for Voice Clone Detection' accepted at AIRC (2025).
- 3rd Place — Procom Computer Vision Competition (medical image segmentation).
- 3rd Place — PaysysLab AI Competition (biometric fingerprint extraction with YOLO v11).

- Docker Training Course for the Absolute Beginner — Certified.
- Dean's List — FAST NUCES, BS Computer Science.

LANGUAGES

English (Professional Working) · Urdu (Native)